KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

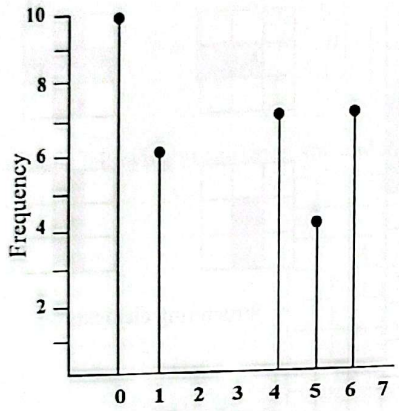
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

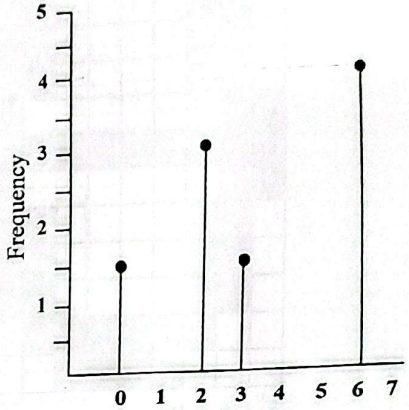
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Construct a modified image using histogram specification. The histograms of original and target images are given below. (20)



(a) Original Image



(a) Target Image

b) Use 8-connectivity and m-connectivity to find the shortest path between pixels (1, 1) with value 10 and (8, 8) with value 100, if V = {0, 10, 50, 65, 100, 106}. (10)

10	55	60	65	05
106	65	56	102	200
112	0	50	50	150
30	45	65	60	120
45	106	10	65	100

c) Let, you have computed the 2D DFT of a given image. The magnitude spectrum ranges from 0 to 10^5, but you want to display the image as an 8-bit gray scale image. For this purpose, you apply base-10 log transformation to the spectrum. What must be the normalizing constant of the transformation? (05)

2. a) Following is a 5×5 image with 5 distinct gray levels: (25)

18	18	4	12	12
10	10	4	12	4
10	4	12	4	15
12	10	15	15	10
15	12	4	10	4

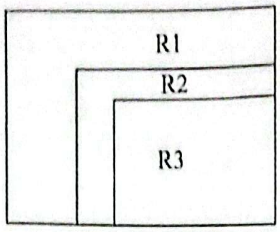
Compare the efficiency of compressing the image using Huffman encoding and Golumb coding with parameter, $m = 5$.

b) Explain why transformation function to perform image complement approximation is different for each component in HSI model but similar for all components in RGB model. (10)

3. a) Consider a line of intensity values, {100, 129, 145, 56, 99} If it is quantized with 4 higher bit planes, compute the error and signal-to-noise ratios for the quantized data. (08)

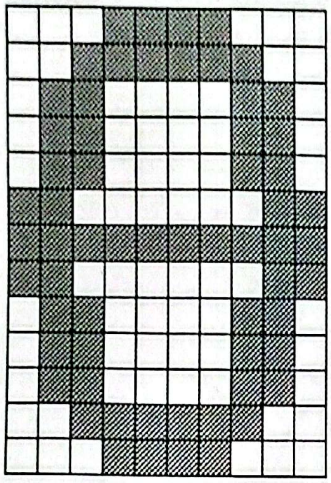
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b) Explain which type of histogram equalization would perform better visual improvement for the following image. (12)

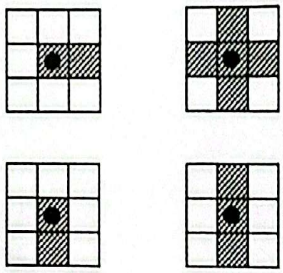


Pixel values of region, R1 = 127, R2 = 129, R3 = 130.

c) Figure below shows a binary image containing Greek letter θ, and four structuring elements. What morphological operations(s) would you use to obtain the number zero (0) from the letter θ? Consider that you can use any of the given structuring elements as many times as you wish. Provide a reason for your choice of operation(s) as well as the structuring elements. (15)

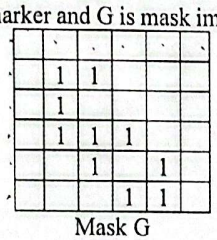


Binary image

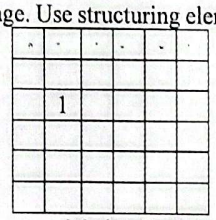


Structuring elements

4. a) Sketch the result of the following operations: (10)
$D_G^{(n)}(F)$ where F is marker and G is mask image. Use structuring element of 3×3 with all 1's.



Mask G



Marker F

b) Use necessary diagrams to show the conceptual relationship between the RGB and HSI color models. (12)

c) Explain with necessary mathematical formulation, how you can apply a linear filter on a color image represented in HSV color space. Also mention the need of saturation weighted hue component. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the property of a linear operation. With necessary mathematical formulation, show that convolution is a linear operation. (10)

b) You are given the following spatial domain filter. (15)

0.15	0.35	0	−0.35	−0.15

i) Compute Fourier spectrum and phase of this filter,

ii) What do you think the effect of applying this filter to an image will be? Justify your answer.

c) Construct a homomorphic filtering function using a Butterworth high pass filter to correct nonuniform illumination. (10)

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6. a) Consider the image histogram given below. (18)

Intensity level	0	1	2	3	4	5
Frequency	14	11	4	12	16	7

Use maximum entropy-based thresholding technique to find a global threshold to segment the image.

b) State the necessity of non-maximum suppression stage used in Canny edge detection algorithm. Provide an idea to get edge orientation by avoiding trigonometric function that can be used in the Canny's algorithm. (12)

c) "Median filter may reduce the number of occupied bins in a histogram." – justify the statement. (05)

7. a) A 100×100 image is warped applying the following affine transformation matrix T. Mention the warped image size. Let, you are currently at pixel (50, 65) in the warped image, find its intensity if inverse mapping technique and bilinear interpolation were used during warping. Consider the following image segment of the input image. (16)

$$T = \begin{bmatrix} 1 & 0.6 & 5 \\ 1.2 & 0.8 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

y \ x	36	37	38	39
10	60	65	70	72
11	66	70	80	88
12	80	90	95	120
13	90	92	110	150

image segment

b) Segment the following image using spit and merged procedure. (12)

	1	1	1	1	1	
	1	1	1	1	1	
	1			1	1	
	1			1	1	

c) Explain aperture problem in optical flow estimation. (07)

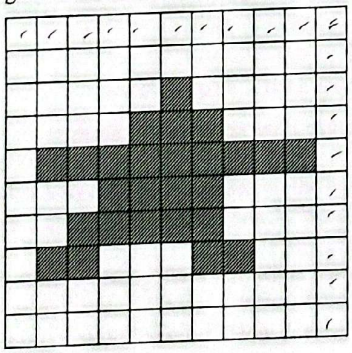
8. a) Let you are going to restore a noisy image having only additive noise. For the restoration purpose you construct the following histogram of a flat region of the image. (12)

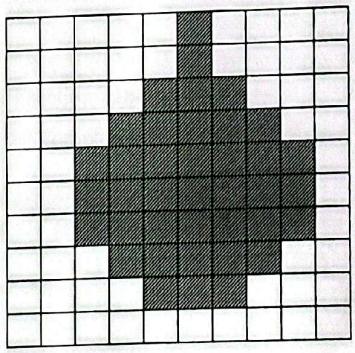
Intensity level	0	1	2	3	4	5	6	7
Probability	0	0	0.210	0.404	0.277	0.102	0.022	0.003

i) Determine the type of noise present,

ii) Provide an estimation of the parameters: mean and standard deviation of the noise density function.

b) Find the region descriptors: form factor, compactness, and circularity to describe the binary region in the following images. (12)





c) Derive the brightness consistency equation to estimate motion field as a form of optical flow. Suggest an idea to overcome its limitation. (11)

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